

2.03-hpi-partialresponses-dev

September 2, 2024

```
[1]: %load_ext autoreload
      %autoreload 2

import numpy as np
import pandas as pd
import torch
from prism.config import PROCESSED_DATA_DIR, MODELS_DIR
from prism.legacy import normalise
from prism.load_models import load_mlp
from prism.pr_save_test import save_partial_responses_results,
    ↪test_refactored_partial_responses, load_partial_responses,
    ↪compare_and_visualize
from prism.partial_responses import partial_responses, PartialResponseCalculator
from prism.lasso import lasso
from prism.nomogram import nomogram
```

```
2024-08-20 11:02:43.118 | INFO      |
prism.config:<module>:11 - PROJ_ROOT path is:
C:\Users\localuser\PRiSM\prism_github
```

```
[2]: def print_first_responses(x_train, x_test, rows=5, columns=5, n_univariate=11):
      print(f"First few univariate responses (train):\n{x_train[:rows, :
      ↪columns]}\n")
      print(f"First few bivariate responses (train):\n{x_train[:rows,
      ↪n_univariate:(n_univariate+columns)]}\n")
      print(f"First few univariate responses (test):\n{x_test[:rows, :columns]}\n")
      print(f"First few bivariate responses (test):\n{x_test[:rows, n_univariate:
      ↪(n_univariate+columns)]}\n")
```

```
[3]: import matplotlib.pyplot as plt
      import seaborn as sns

def find_large_deviations(torch_tensor, numpy_array, threshold=0.1):
    diff = np.abs(torch_tensor.numpy() - numpy_array)
    large_dev_indices = np.argwhere(diff > threshold)
    return large_dev_indices
```

```

def visualize_large_deviations(torch_tensor, numpy_array, large_dev_indices,
    num_samples=5):
    fig, axes = plt.subplots(num_samples, 3, figsize=(15, 5*num_samples))

    for i, (row, col) in enumerate(large_dev_indices[:num_samples]):
        window_size = 5
        row_start, row_end = max(0, row-window_size), min(torch_tensor.
            shape[0], row+window_size+1)
        col_start, col_end = max(0, col-window_size), min(torch_tensor.
            shape[1], col+window_size+1)

        torch_subset = torch_tensor[row_start:row_end, col_start:col_end].
            numpy()
        numpy_subset = numpy_array[row_start:row_end, col_start:col_end]
        diff_subset = np.abs(torch_subset - numpy_subset)

        vmin = min(torch_subset.min(), numpy_subset.min())
        vmax = max(torch_subset.max(), numpy_subset.max())

        axes[i, 0].imshow(torch_subset, cmap='coolwarm', vmin=vmin, vmax=vmax)
        axes[i, 0].set_title(f'PyTorch (Row: {row}, Col: {col})')
        axes[i, 1].imshow(numpy_subset, cmap='coolwarm', vmin=vmin, vmax=vmax)
        axes[i, 1].set_title(f'NumPy (Row: {row}, Col: {col})')
        im = axes[i, 2].imshow(diff_subset, cmap='viridis')
        axes[i, 2].set_title('Absolute Difference')
        plt.colorbar(im, ax=axes[i, 2])

    plt.tight_layout()
    plt.show()

def scatter_plot_large_deviations(large_dev_indices, shape):
    plt.figure(figsize=(12, 8))
    plt.scatter(large_dev_indices[:, 0], large_dev_indices[:, 1], alpha=0.5)
    plt.xlabel('Row Index')
    plt.ylabel('Column Index')
    plt.title('Distribution of Large Deviations')
    plt.xlim(0, shape[0])
    plt.ylim(0, shape[1])
    plt.gca().invert_yaxis() # Invert y-axis to match matrix coordinates
    plt.colorbar(label='Density')
    plt.show()

def compare_and_visualize(torch_tensor, numpy_array):
    print("Data Types:")
    print(f"PyTorch Tensor: {torch_tensor.dtype}")
    print(f"NumPy Array: {numpy_array.dtype}")
    print()

```

```

print("Shape:")
print(f"PyTorch Tensor: {torch_tensor.shape}")
print(f"NumPy Array: {numpy_array.shape}")
print()

print("Basic Statistics:")
print("Mean:")
print(f"PyTorch Tensor: {torch_tensor.mean().item()}")
print(f"NumPy Array: {numpy_array.mean()}")
print()

print("Standard Deviation:")
print(f"PyTorch Tensor: {torch_tensor.std().item()}")
print(f"NumPy Array: {numpy_array.std()}")
print()

print("Min:")
print(f"PyTorch Tensor: {torch_tensor.min().item()}")
print(f"NumPy Array: {numpy_array.min()}")
print()

print("Max:")
print(f"PyTorch Tensor: {torch_tensor.max().item()}")
print(f"NumPy Array: {numpy_array.max()}")
print()

# Overall distribution plot
plt.figure(figsize=(12, 6))
sns.histplot(torch_tensor.numpy().flatten(), kde=True, color='blue',
↪alpha=0.5, label='PyTorch')
sns.histplot(numpy_array.flatten(), kde=True, color='red', alpha=0.5,
↪label='NumPy')
plt.title('Distribution of Values')
plt.legend()
plt.show()

# Correlation plot
plt.figure(figsize=(10, 8))
plt.scatter(torch_tensor.numpy().flatten(), numpy_array.flatten(), alpha=0.
↪1)
plt.xlabel('PyTorch Tensor Values')
plt.ylabel('NumPy Array Values')
plt.title('Correlation between PyTorch Tensor and NumPy Array')
plt.plot([numpy_array.min(), numpy_array.max()], [numpy_array.min(),
↪numpy_array.max()], 'r--', lw=2)
plt.show()

```

```

# Find and visualize large deviations
large_dev_indices = find_large_deviations(torch_tensor, numpy_array)
visualize_large_deviations(torch_tensor, numpy_array, large_dev_indices)

# Scatter plot of large deviation indices
scatter_plot_large_deviations(large_dev_indices, torch_tensor.shape)

# Print summary of large deviations
print(f"Number of elements with large deviations: {len(large_dev_indices)}")
print(f"Percentage of elements with large deviations:␣
↪{len(large_dev_indices) / torch_tensor.numel() * 100:.2f}%")

```

1 Load and setup

```
[4]: %reload_ext autoreload
```

```
[5]: seed = 257
np.random.seed(seed)
torch.manual_seed(seed)
```

```
[5]: <torch._C.Generator at 0x1af7ad8afb0>
```

```
[6]: import matplotlib.pyplot as plt

def plot_histograms(pr_train_benchmark, x_train, num_subplot_cols=3):
    # Univariate responses only
    m = x_train.shape[1]

    # Calculate number of rows needed for the subplots
    num_subplot_rows = (m + num_subplot_cols - 1) // num_subplot_cols

    # Create a figure with subplots arranged in 3 columns
    fig, axes = plt.subplots(num_subplot_rows, num_subplot_cols, figsize=(12, 4␣
↪* num_subplot_rows))
    fig.suptitle(f'Histograms of Benchmark Training Partial Responses')

    # Plot histogram for each column
    for i in range(m):
        row = i // num_subplot_cols
        col = i % num_subplot_cols
        ax = axes[row, col] if num_subplot_rows > 1 else axes[col]
        ax.hist(pr_train_benchmark[:, i], bins=15, edgecolor='black')
        ax.set_title(x_train.columns[i])
        ax.set_xlabel('Value')
        ax.set_ylabel('Frequency')
```

```

# Remove any unused subplots
for i in range(m, num_subplot_rows * num_subplot_cols):
    row = i // num_subplot_cols
    col = i % num_subplot_cols
    fig.delaxes(axes[row, col] if num_subplot_rows > 1 else axes[col])

# Adjust layout and show the plot
plt.tight_layout()
plt.show()

```

1.1 Load and preprocess data

```

[7]: data_train = pd.read_csv(PROCESSED_DATA_DIR.joinpath('imputed_dataset1_train.
    ↪csv'))
data_test = pd.read_csv(PROCESSED_DATA_DIR.joinpath('imputed_dataset1_test.
    ↪csv'))
data_val = pd.read_csv(PROCESSED_DATA_DIR.joinpath('imputed_dataset1_val.csv'))

data_train_test = pd.concat([data_train, data_test]) # used for axis annotation,
    ↪in some plotting functions

# drop id column
data_train.drop('trr_id_code', axis=1, inplace=True)
data_test.drop('trr_id_code', axis=1, inplace=True)
data_val.drop('trr_id_code', axis=1, inplace=True)

target_col = 'oneyearmort'

x_train0 = data_train.drop(target_col, axis=1)
y_train = data_train[target_col]

x_test0 = data_test.drop(target_col, axis=1)
y_test = data_test[target_col]

x_val0 = data_val.drop(target_col, axis=1)
y_val = data_val[target_col]

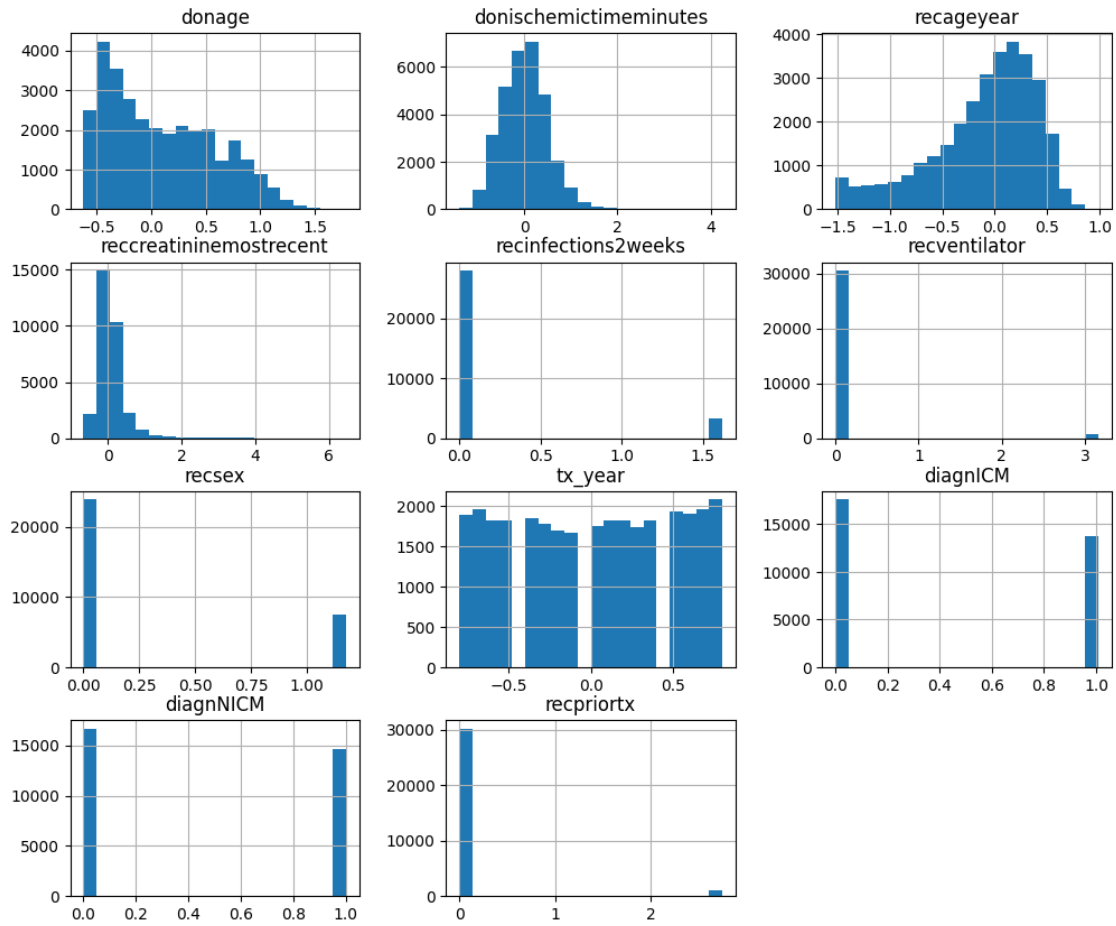
[x_train, x_test] = normalise(x_train0, x_test0)
x_val = normalise(x_val0)

```

```

[8]: x_train.hist(bins=20, figsize=(12, 10));

```



```
[9]: feature_names = [
    'don age',
    'don isch time min',
    'rec age yr',
    'rec creat',
    'rec infect 2wk',
    'rec vent',
    'rec sex',
    'tx year',
    'ICM',
    'NICM',
    'prior tx'
]
```

1.2 Load MLP

```
[10]: filename_mlp = 'mlp_model_20240705_135534'
try:
    mlp, mlp_params, mlp_metrics = load_mlp(filename_mlp, MODELS_DIR)
    print("Model loaded successfully")
    print("Derived Model Structure:")
    print("\nModel Parameters:")
    print(mlp_params)
    print("\nModel Metrics:")
    print(mlp_metrics)
except Exception as e:
    print(f"Error loading model: {e}")
    raise

# Extract method and device from mlp_params, with defaults
method = mlp_params.get('method', 'dirac')
device = mlp_params.get('device', 'cpu')
```

Model loaded successfully

Derived Model Structure:

Model Parameters:

```
{'n_hidden': 10, 'weight_decay': 1e-05, 'lr': 0.001, 'patience': 50,
'tolerance': 0.0001, 'batch_size': 1024, 'device': 'cpu', 'seed': 257}
```

Model Metrics:

```
{'prevalence': 0.123, 'sensitivity': 0.241, 'specificity': 0.864, 'accuracy':
0.797, 'ppv': 0.175, 'auc score': 0.601, 'auc lower ci': '0.577', 'auc upper
ci': '0.626'}
```

1.3 Load benchmark partial responses

```
[11]: pr_load = load_partial_responses()
pr_train_benchmark = pr_load['partial_responses_train']
pr_test_benchmark = pr_load['partial_responses_test']
```

Data successfully loaded from

C:\Users\localuser\PRiSM\prism_github\models\partial_responses_data.pkl

Loaded data contains the following keys: x_train, x_test, model,

partial_responses_train, partial_responses_test, method, device

x_train shape: (31315, 11)

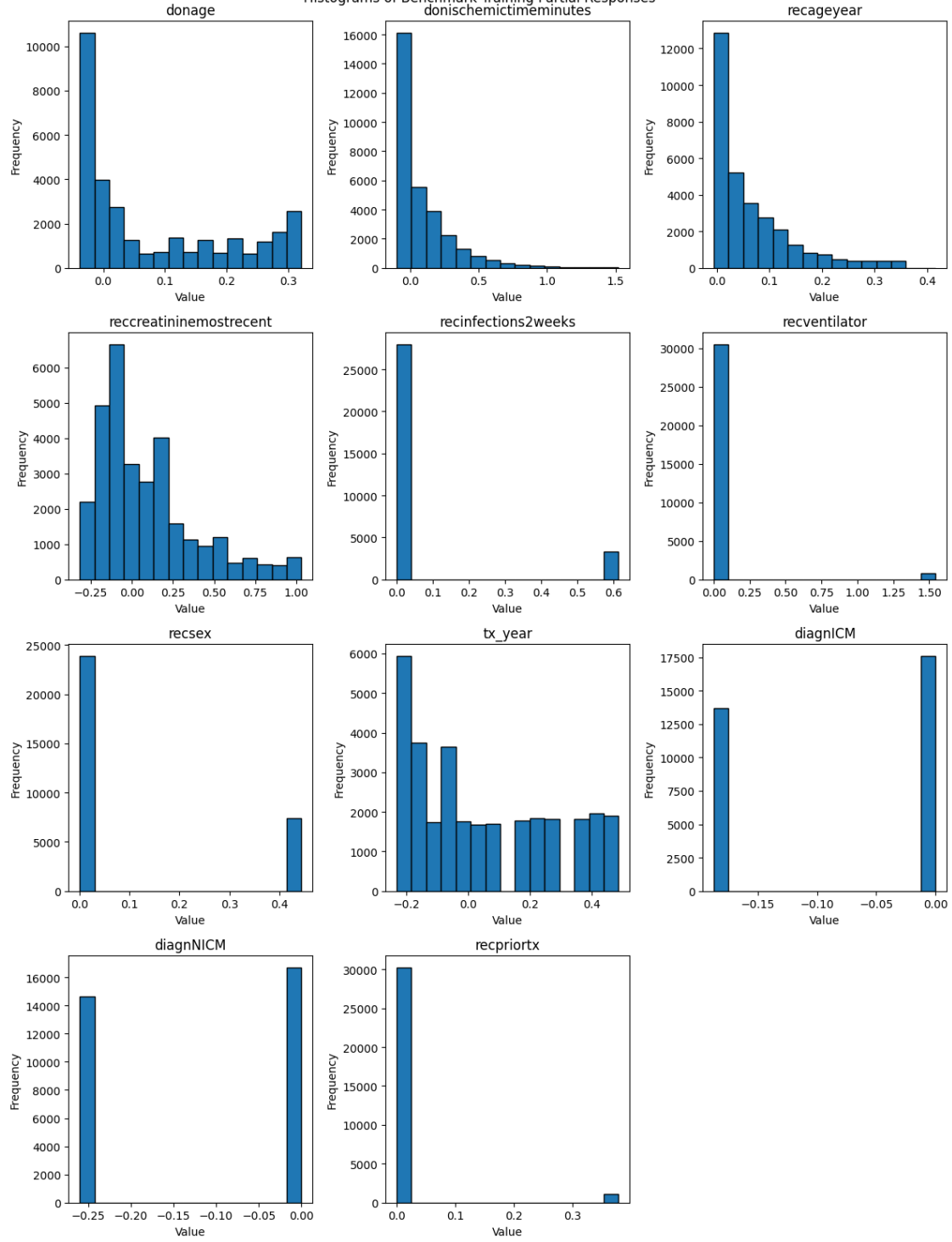
x_test shape: (6120, 11)

partial_responses_train shape: (31315, 66)

partial_responses_test shape: (6120, 66)

```
[12]: plot_histograms(pr_train_benchmark, x_train)
```

Histograms of Benchmark Training Partial Responses



2 Test Dirac

```
[13]: x_train_tensor = torch.tensor(x_train.values, dtype=torch.float32,
    ↪device=device)
    x_test_tensor = torch.tensor(x_test.values, dtype=torch.float32, device=device)

[14]: pr = PartialResponseCalculator(mlp, method, device, input_dim=x_train_tensor.
    ↪shape[1])

[15]: univariate_train, bivariate_train, bivariate_inputs = pr.
    ↪calculate(x_train_tensor)

[16]: pr_train, pr_test, bivariate_inputs =
    ↪partial_responses(x_train_tensor, x_test_tensor, mlp, method='dirac', device=device)

[17]: print_first_responses(pr_train, pr_test)
```

First few univariate responses (train):

```
tensor([[ 0.0716, -0.0867, -0.0057,  0.2100,  0.0000],
        [-0.0269,  0.1127, -0.0032, -0.1168,  0.0000],
        [ 0.2512, -0.0407,  0.0694,  0.0664,  0.0000],
        [ 0.2083,  0.6771,  0.0565,  0.4328,  0.0000],
        [-0.0382, -0.0836,  0.0169, -0.2702,  0.0000]])
```

First few bivariate responses (train):

```
tensor([[ 0.0065, -0.0075, -0.0054,  0.0000,  0.0000],
        [ 0.0166,  0.0041, -0.0159,  0.0000,  0.0000],
        [-0.0001,  0.0402, -0.0058,  0.0000,  0.0000],
        [-0.0387,  0.0310, -0.0015,  0.0000,  0.0000],
        [-0.0411, -0.0092, -0.0393,  0.0000,  0.0000]])
```

First few univariate responses (test):

```
tensor([[ -0.0102,  0.1832,  0.0044,  0.0664,  0.0000],
        [-0.0382,  0.0829,  0.2849,  0.1513,  0.0000],
        [-0.0326,  0.2240,  0.0342,  0.0000,  0.0000],
        [-0.0269, -0.0407,  0.1513, -0.0435,  0.0000],
        [ 0.3009,  0.1306,  0.1152,  0.5716,  0.0000]])
```

First few bivariate responses (test):

```
tensor([[ 0.0281, -0.0050,  0.0068,  0.0000,  0.0000],
        [ 0.0078,  0.0090,  0.0065,  0.0000,  0.0000],
        [ 0.0256, -0.0202,  0.0000,  0.0000,  0.0000],
        [-0.0090, -0.0612, -0.0050,  0.0000,  0.0000],
        [ 0.0016,  0.0615,  0.0329,  0.0000,  0.0000]])
```

```
[18]: print_first_responses(pr_train_benchmark, pr_test_benchmark)
```

First few univariate responses (train):

```
[[ 0.07161093 -0.08673906 -0.00569773  0.20996237  0.          ]
 [-0.02694654  0.11274576 -0.00319147 -0.11675644  0.          ]
 [ 0.25121975 -0.04073429  0.06940937  0.06640053  0.          ]
 [ 0.20826769  0.67714942  0.05648136  0.43277049  0.          ]
```

```

[-0.03824949 -0.08359814  0.01686096 -0.27024698  0.          ]]
First few bivariate responses (train):
[[ 0.00654554 -0.00751519 -0.00542164  0.          0.          ]
 [ 0.01660681  0.00406265 -0.01590681  0.          0.          ]
 [-0.00010848  0.04021502 -0.00579095  0.          0.          ]
 [-0.03867877  0.03100348 -0.00154138  0.          0.          ]
 [-0.04112434 -0.00924683 -0.03928757  0.          0.          ]]
First few univariate responses (test):
[[-0.01015663  0.1832056  0.0043993  0.06640053  0.          ]
 [-0.03822756  0.08289409  0.28486729  0.15125036  0.          ]
 [-0.03259659  0.22404289  0.03423142  0.          0.          ]
 [-0.02694654 -0.04073429  0.15129423 -0.04352379  0.          ]
 [ 0.3008616  0.13062477  0.11519027  0.57161987  0.          ]]
First few bivariate responses (test):
[[ 0.02811289 -0.00499439  0.00679827  0.          0.          ]
 [ 0.00776172  0.00899982  0.00646639  0.          0.          ]
 [ 0.02556467 -0.02021384  0.          0.          0.          ]
 [-0.00898981 -0.06119585 -0.0049758  0.          0.          ]
 [ 0.00155044  0.06150293  0.03290844  0.          0.          ]]

```

2.1 compare dirac results

```

[19]: ## test dirac
      test_refactored_partial_responses(partial_responses, tensor_input=True)

```

Refactored function produces different results from the original implementation.
atol = 1e-07, rtol = 1e-05
Differences found in partial_responses_train
Max absolute difference: 1.430511474609375e-06
Max relative difference: 0.00010408634670246127
Differences found in partial_responses_test
Max absolute difference: 1.430511474609375e-06
Max relative difference: 9.475296008247299e-05

```

[19]: False

```

3 Test Lebesgue CPU 500

```

[20]: # Load results from old method
      pr_load_leb= load_partial_responses(MODELS_DIR.
      ↪joinpath("partial_responses_data_leb500.pkl"))
      pr_train_benchmark_leb = pr_load_leb['partial_responses_train']
      pr_test_benchmark_leb = pr_load_leb['partial_responses_test']

```

Data successfully loaded from
C:\Users\localuser\PRiSM\prism_github\models\partial_responses_data_leb500.pkl
Loaded data contains the following keys: x_train, x_test, model,
partial_responses_train, partial_responses_test, method, device

```

x_train shape: (500, 11)
x_test shape: (500, 11)
partial_responses_train shape: (500, 66)
partial_responses_test shape: (500, 66)

```

```
[21]: print_first_responses(pr_train_benchmark_leb, pr_test_benchmark_leb)
```

```

First few univariate responses (train):
[[ 0.02279246 -0.13538277 -0.08238709  0.05137408 -0.04876649]
 [-0.18926609  0.06411755 -0.04966962 -0.16132271 -0.04876649]
 [ 0.24780893 -0.07403123  0.07245362 -0.04652226 -0.04876649]
 [ 0.1899035   0.45994484  0.0572989   0.2237339  -0.04876649]
 [-0.1667074  -0.20167506  0.00046611 -0.25231373 -0.04876649]]

First few bivariate responses (train):
[[ 0.00975614 -0.0014504 -0.00784853  0.0015941 -0.00162662]
 [ 0.00449143 -0.00350396 -0.01813054 -0.00167009 -0.00617562]
 [ 0.00969791  0.01454909 -0.00556087  0.0005069   0.00979852]
 [-0.05770344  0.01386476 -0.0322869   0.00121743  0.00653278]
 [-0.0161572  -0.00503564 -0.03246517 -0.00096593 -0.00687471]]

First few univariate responses (test):
[[-0.53211224 -0.39187014 -0.45838583 -0.41380584 -0.44975626]
 [-0.54525983 -0.43861377 -0.47356451 -0.33227766 -0.44975626]
 [-0.54459131 -0.37141097 -0.40668476 -0.47929466 -0.44975626]
 [-0.54169858 -0.48624599 -0.27472937 -0.52340114 -0.44975626]
 [-0.11911523 -0.41703665 -0.31105554  0.06441379 -0.44975626]]

First few bivariate responses (test):
[[0.42502758 0.40714574 0.42756259 0.41155931 0.41243818]
 [0.41156316 0.49267107 0.42399994 0.41126719 0.41159272]
 [0.42337883 0.39622626 0.41832441 0.41132978 0.41194433]
 [0.40220404 0.36848888 0.41442394 0.41138661 0.41209343]
 [0.41623333 0.46823516 0.39210296 0.40820089 0.41403693]]

```

```

[22]: # View new lebesgue results
%reload_ext autoreload

pr_train_leb, pr_test_leb, bivariate_inputs_leb = 
↳ partial_responses(x_train_tensor[:500,:], x_test_tensor[:500,:
↳ ], mlp, method='lebesgue', device='cpu')

```

```

Main compute device: cpu
Max threads: 15.
Univariate 0, batch_start 0      (cpu)
Univariate 1, batch_start 0      (cpu)
Univariate 2, batch_start 0      (cpu)
Univariate 3, batch_start 0      (cpu)
Univariate 4, batch_start 0      (cpu)
Univariate 5, batch_start 0      (cpu)
Univariate 6, batch_start 0      (cpu)
Univariate 7, batch_start 0      (cpu)

```

Univariate 8, batch_start 0 (cpu)
Univariate 9, batch_start 0 (cpu)
Univariate 10, batch_start 0 (cpu)
Bivariate 0,1, batch_start 0 (cpu)
Bivariate 0,2, batch_start 0 (cpu)
Bivariate 0,3, batch_start 0 (cpu)
Bivariate 0,4, batch_start 0 (cpu)
Bivariate 0,5, batch_start 0 (cpu)
Bivariate 0,6, batch_start 0 (cpu)
Bivariate 0,7, batch_start 0 (cpu)
Bivariate 0,8, batch_start 0 (cpu)
Bivariate 0,9, batch_start 0 (cpu)
Bivariate 0,10, batch_start 0 (cpu)
Bivariate 1,2, batch_start 0 (cpu)
Bivariate 1,3, batch_start 0 (cpu)
Bivariate 1,4, batch_start 0 (cpu)
Bivariate 1,5, batch_start 0 (cpu)
Bivariate 1,6, batch_start 0 (cpu)
Bivariate 1,7, batch_start 0 (cpu)
Bivariate 1,8, batch_start 0 (cpu)
Bivariate 1,9, batch_start 0 (cpu)
Bivariate 1,10, batch_start 0 (cpu)
Bivariate 2,3, batch_start 0 (cpu)
Bivariate 2,4, batch_start 0 (cpu)
Bivariate 2,5, batch_start 0 (cpu)
Bivariate 2,6, batch_start 0 (cpu)
Bivariate 2,7, batch_start 0 (cpu)
Bivariate 2,8, batch_start 0 (cpu)
Bivariate 2,9, batch_start 0 (cpu)
Bivariate 2,10, batch_start 0 (cpu)
Bivariate 3,4, batch_start 0 (cpu)
Bivariate 3,5, batch_start 0 (cpu)
Bivariate 3,6, batch_start 0 (cpu)
Bivariate 3,7, batch_start 0 (cpu)
Bivariate 3,8, batch_start 0 (cpu)
Bivariate 3,9, batch_start 0 (cpu)
Bivariate 3,10, batch_start 0 (cpu)
Bivariate 4,5, batch_start 0 (cpu)
Bivariate 4,6, batch_start 0 (cpu)
Bivariate 4,7, batch_start 0 (cpu)
Bivariate 4,8, batch_start 0 (cpu)
Bivariate 4,9, batch_start 0 (cpu)
Bivariate 4,10, batch_start 0 (cpu)
Bivariate 5,6, batch_start 0 (cpu)
Bivariate 5,7, batch_start 0 (cpu)
Bivariate 5,8, batch_start 0 (cpu)
Bivariate 5,9, batch_start 0 (cpu)
Bivariate 5,10, batch_start 0 (cpu)

```

Bivariate 6,7, batch_start 0    (cpu)
Bivariate 6,8, batch_start 0    (cpu)
Bivariate 6,9, batch_start 0    (cpu)
Bivariate 6,10, batch_start 0   (cpu)
Bivariate 7,8, batch_start 0    (cpu)
Bivariate 7,9, batch_start 0    (cpu)
Bivariate 7,10, batch_start 0   (cpu)
Bivariate 8,9, batch_start 0    (cpu)
Bivariate 8,10, batch_start 0   (cpu)
Bivariate 9,10, batch_start 0   (cpu)
Main compute device: cpu
Max threads: 15.
Univariate 0, batch_start 0     (cpu)
Univariate 1, batch_start 0     (cpu)
Univariate 2, batch_start 0     (cpu)
Univariate 3, batch_start 0     (cpu)
Univariate 4, batch_start 0     (cpu)
Univariate 5, batch_start 0     (cpu)
Univariate 6, batch_start 0     (cpu)
Univariate 7, batch_start 0     (cpu)
Univariate 8, batch_start 0     (cpu)
Univariate 9, batch_start 0     (cpu)
Univariate 10, batch_start 0    (cpu)
Bivariate 0,1, batch_start 0    (cpu)
Bivariate 0,2, batch_start 0    (cpu)
Bivariate 0,3, batch_start 0    (cpu)
Bivariate 0,4, batch_start 0    (cpu)
Bivariate 0,5, batch_start 0    (cpu)
Bivariate 0,6, batch_start 0    (cpu)
Bivariate 0,7, batch_start 0    (cpu)
Bivariate 0,8, batch_start 0    (cpu)
Bivariate 0,9, batch_start 0    (cpu)
Bivariate 0,10, batch_start 0   (cpu)
Bivariate 1,2, batch_start 0    (cpu)
Bivariate 1,3, batch_start 0    (cpu)
Bivariate 1,4, batch_start 0    (cpu)
Bivariate 1,5, batch_start 0    (cpu)
Bivariate 1,6, batch_start 0    (cpu)
Bivariate 1,7, batch_start 0    (cpu)
Bivariate 1,8, batch_start 0    (cpu)
Bivariate 1,9, batch_start 0    (cpu)
Bivariate 1,10, batch_start 0   (cpu)
Bivariate 2,3, batch_start 0    (cpu)
Bivariate 2,4, batch_start 0    (cpu)
Bivariate 2,5, batch_start 0    (cpu)
Bivariate 2,6, batch_start 0    (cpu)
Bivariate 2,7, batch_start 0    (cpu)
Bivariate 2,8, batch_start 0    (cpu)

```

```

Bivariate 2,9, batch_start 0    (cpu)
Bivariate 2,10, batch_start 0   (cpu)
Bivariate 3,4, batch_start 0    (cpu)
Bivariate 3,5, batch_start 0    (cpu)
Bivariate 3,6, batch_start 0    (cpu)
Bivariate 3,7, batch_start 0    (cpu)
Bivariate 3,8, batch_start 0    (cpu)
Bivariate 3,9, batch_start 0    (cpu)
Bivariate 3,10, batch_start 0   (cpu)
Bivariate 4,5, batch_start 0    (cpu)
Bivariate 4,6, batch_start 0    (cpu)
Bivariate 4,7, batch_start 0    (cpu)
Bivariate 4,8, batch_start 0    (cpu)
Bivariate 4,9, batch_start 0    (cpu)
Bivariate 4,10, batch_start 0   (cpu)
Bivariate 5,6, batch_start 0    (cpu)
Bivariate 5,7, batch_start 0    (cpu)
Bivariate 5,8, batch_start 0    (cpu)
Bivariate 5,9, batch_start 0    (cpu)
Bivariate 5,10, batch_start 0   (cpu)
Bivariate 6,7, batch_start 0    (cpu)
Bivariate 6,8, batch_start 0    (cpu)
Bivariate 6,9, batch_start 0    (cpu)
Bivariate 6,10, batch_start 0   (cpu)
Bivariate 7,8, batch_start 0    (cpu)
Bivariate 7,9, batch_start 0    (cpu)
Bivariate 7,10, batch_start 0   (cpu)
Bivariate 8,9, batch_start 0    (cpu)
Bivariate 8,10, batch_start 0   (cpu)
Bivariate 9,10, batch_start 0   (cpu)

```

```
[23]: print_first_responses(pr_train_leb, pr_test_leb)
```

First few univariate responses (train):

```

tensor([[ 0.0228, -0.1354, -0.0824,  0.0514, -0.0488],
        [-0.1893,  0.0641, -0.0497, -0.1613, -0.0488],
        [ 0.2478, -0.0740,  0.0725, -0.0465, -0.0488],
        [ 0.1899,  0.4599,  0.0573,  0.2237, -0.0488],
        [-0.1667, -0.2017,  0.0005, -0.2523, -0.0488]])

```

First few bivariate responses (train):

```

tensor([[ 0.0098, -0.0015, -0.0078,  0.0016, -0.0016],
        [ 0.0045, -0.0035, -0.0181, -0.0017, -0.0062],
        [ 0.0097,  0.0145, -0.0056,  0.0005,  0.0098],
        [-0.0577,  0.0139, -0.0323,  0.0012,  0.0065],
        [-0.0162, -0.0050, -0.0325, -0.0010, -0.0069]])

```

First few univariate responses (test):

```

tensor([[ -0.5321, -0.3919, -0.4584, -0.4138, -0.4498],
        [ -0.5453, -0.4386, -0.4736, -0.3323, -0.4498],

```

```

        [-0.5446, -0.3714, -0.4067, -0.4793, -0.4498],
        [-0.5417, -0.4862, -0.2747, -0.5234, -0.4498],
        [-0.1191, -0.4170, -0.3111, 0.0644, -0.4498]])
First few bivariate responses (test):
tensor([[0.4250, 0.4071, 0.4276, 0.4116, 0.4124],
        [0.4116, 0.4927, 0.4240, 0.4113, 0.4116],
        [0.4234, 0.3962, 0.4183, 0.4113, 0.4119],
        [0.4022, 0.3685, 0.4144, 0.4114, 0.4121],
        [0.4162, 0.4682, 0.3921, 0.4082, 0.4140]])

```

```

[24]: # Test lebesgue
      %reload_ext autoreload
      test_refactored_partial_responses(partial_responses,filename=MODELS_DIR.
      ↪joinpath("partial_responses_data_leb500.pkl"), tensor_input=True)

```

```

Main compute device: cpu
Max threads: 15.
Univariate 0, batch_start 0      (cpu)
Univariate 1, batch_start 0      (cpu)
Univariate 2, batch_start 0      (cpu)
Univariate 3, batch_start 0      (cpu)
Univariate 4, batch_start 0      (cpu)
Univariate 5, batch_start 0      (cpu)
Univariate 6, batch_start 0      (cpu)
Univariate 7, batch_start 0      (cpu)
Univariate 8, batch_start 0      (cpu)
Univariate 9, batch_start 0      (cpu)
Univariate 10, batch_start 0     (cpu)
Bivariate 0,1, batch_start 0     (cpu)
Bivariate 0,2, batch_start 0     (cpu)
Bivariate 0,3, batch_start 0     (cpu)
Bivariate 0,4, batch_start 0     (cpu)
Bivariate 0,5, batch_start 0     (cpu)
Bivariate 0,6, batch_start 0     (cpu)
Bivariate 0,7, batch_start 0     (cpu)
Bivariate 0,8, batch_start 0     (cpu)
Bivariate 0,9, batch_start 0     (cpu)
Bivariate 0,10, batch_start 0    (cpu)
Bivariate 1,2, batch_start 0     (cpu)
Bivariate 1,3, batch_start 0     (cpu)
Bivariate 1,4, batch_start 0     (cpu)
Bivariate 1,5, batch_start 0     (cpu)
Bivariate 1,6, batch_start 0     (cpu)
Bivariate 1,7, batch_start 0     (cpu)
Bivariate 1,8, batch_start 0     (cpu)
Bivariate 1,9, batch_start 0     (cpu)
Bivariate 1,10, batch_start 0    (cpu)
Bivariate 2,3, batch_start 0     (cpu)

```

```

Bivariate 2,4, batch_start 0    (cpu)
Bivariate 2,5, batch_start 0    (cpu)
Bivariate 2,6, batch_start 0    (cpu)
Bivariate 2,7, batch_start 0    (cpu)
Bivariate 2,8, batch_start 0    (cpu)
Bivariate 2,9, batch_start 0    (cpu)
Bivariate 2,10, batch_start 0   (cpu)
Bivariate 3,4, batch_start 0    (cpu)
Bivariate 3,5, batch_start 0    (cpu)
Bivariate 3,6, batch_start 0    (cpu)
Bivariate 3,7, batch_start 0    (cpu)
Bivariate 3,8, batch_start 0    (cpu)
Bivariate 3,9, batch_start 0    (cpu)
Bivariate 3,10, batch_start 0   (cpu)
Bivariate 4,5, batch_start 0    (cpu)
Bivariate 4,6, batch_start 0    (cpu)
Bivariate 4,7, batch_start 0    (cpu)
Bivariate 4,8, batch_start 0    (cpu)
Bivariate 4,9, batch_start 0    (cpu)
Bivariate 4,10, batch_start 0   (cpu)
Bivariate 5,6, batch_start 0    (cpu)
Bivariate 5,7, batch_start 0    (cpu)
Bivariate 5,8, batch_start 0    (cpu)
Bivariate 5,9, batch_start 0    (cpu)
Bivariate 5,10, batch_start 0   (cpu)
Bivariate 6,7, batch_start 0    (cpu)
Bivariate 6,8, batch_start 0    (cpu)
Bivariate 6,9, batch_start 0    (cpu)
Bivariate 6,10, batch_start 0   (cpu)
Bivariate 7,8, batch_start 0    (cpu)
Bivariate 7,9, batch_start 0    (cpu)
Bivariate 7,10, batch_start 0   (cpu)
Bivariate 8,9, batch_start 0    (cpu)
Bivariate 8,10, batch_start 0   (cpu)
Bivariate 9,10, batch_start 0   (cpu)
Main compute device: cpu
Max threads: 15.
Univariate 0, batch_start 0    (cpu)
Univariate 1, batch_start 0    (cpu)
Univariate 2, batch_start 0    (cpu)
Univariate 3, batch_start 0    (cpu)
Univariate 4, batch_start 0    (cpu)
Univariate 5, batch_start 0    (cpu)
Univariate 6, batch_start 0    (cpu)
Univariate 7, batch_start 0    (cpu)
Univariate 8, batch_start 0    (cpu)
Univariate 9, batch_start 0    (cpu)
Univariate 10, batch_start 0   (cpu)

```


Bivariate 0,1, batch_start 0 (cpu)
Bivariate 0,2, batch_start 0 (cpu)
Bivariate 0,3, batch_start 0 (cpu)
Bivariate 0,4, batch_start 0 (cpu)
Bivariate 0,5, batch_start 0 (cpu)
Bivariate 0,6, batch_start 0 (cpu)
Bivariate 0,7, batch_start 0 (cpu)
Bivariate 0,8, batch_start 0 (cpu)
Bivariate 0,9, batch_start 0 (cpu)
Bivariate 0,10, batch_start 0 (cpu)
Bivariate 1,2, batch_start 0 (cpu)
Bivariate 1,3, batch_start 0 (cpu)
Bivariate 1,4, batch_start 0 (cpu)
Bivariate 1,5, batch_start 0 (cpu)
Bivariate 1,6, batch_start 0 (cpu)
Bivariate 1,7, batch_start 0 (cpu)
Bivariate 1,8, batch_start 0 (cpu)
Bivariate 1,9, batch_start 0 (cpu)
Bivariate 1,10, batch_start 0 (cpu)
Bivariate 2,3, batch_start 0 (cpu)
Bivariate 2,4, batch_start 0 (cpu)
Bivariate 2,5, batch_start 0 (cpu)
Bivariate 2,6, batch_start 0 (cpu)
Bivariate 2,7, batch_start 0 (cpu)
Bivariate 2,8, batch_start 0 (cpu)
Bivariate 2,9, batch_start 0 (cpu)
Bivariate 2,10, batch_start 0 (cpu)
Bivariate 3,4, batch_start 0 (cpu)
Bivariate 3,5, batch_start 0 (cpu)
Bivariate 3,6, batch_start 0 (cpu)
Bivariate 3,7, batch_start 0 (cpu)
Bivariate 3,8, batch_start 0 (cpu)
Bivariate 3,9, batch_start 0 (cpu)
Bivariate 3,10, batch_start 0 (cpu)
Bivariate 4,5, batch_start 0 (cpu)
Bivariate 4,6, batch_start 0 (cpu)
Bivariate 4,7, batch_start 0 (cpu)
Bivariate 4,8, batch_start 0 (cpu)
Bivariate 4,9, batch_start 0 (cpu)
Bivariate 4,10, batch_start 0 (cpu)
Bivariate 5,6, batch_start 0 (cpu)
Bivariate 5,7, batch_start 0 (cpu)
Bivariate 5,8, batch_start 0 (cpu)
Bivariate 5,9, batch_start 0 (cpu)
Bivariate 5,10, batch_start 0 (cpu)
Bivariate 6,7, batch_start 0 (cpu)
Bivariate 6,8, batch_start 0 (cpu)
Bivariate 6,9, batch_start 0 (cpu)

```

Bivariate 6,10, batch_start 0    (cpu)
Bivariate 7,8, batch_start 0    (cpu)
Bivariate 7,9, batch_start 0    (cpu)
Bivariate 7,10, batch_start 0    (cpu)
Bivariate 8,9, batch_start 0    (cpu)
Bivariate 8,10, batch_start 0    (cpu)
Bivariate 9,10, batch_start 0    (cpu)
Refactored function produces different results from the original implementation.
atol = 1e-07, rtol = 1e-05
Differences found in partial_responses_train
Max absolute difference: 9.092036634683609e-07
Max relative difference: 7.447196628433573e-05
Differences found in partial_responses_test
Max absolute difference: 9.5367431640625e-07
Max relative difference: 3.3892147053959016e-05

```

[24]: False

4 Test lebesgue CPU 2000

```

[25]: import psutil
import os

def get_system_resources():
    # CPU information
    cpu_count = os.cpu_count()
    cpu_usage = psutil.cpu_percent(interval=1)

    # Memory information
    mem = psutil.virtual_memory()
    available_memory = mem.available / (1024 * 1024) # Convert to MB

    print(f"CPU Cores: {cpu_count}")
    print(f"CPU Usage: {cpu_usage}%")
    print(f"Available Memory: {available_memory:.2f} MB")

    return {
        "cpu_count": cpu_count,
        "cpu_usage": cpu_usage,
        "available_memory": available_memory,
    }

resources = get_system_resources()

```

```

CPU Cores: 16
CPU Usage: 20.1%
Available Memory: 14310.73 MB

```

```
[26]: # Load results from old method
pr_load_leb= load_partial_responses(MODELS_DIR.
    ↪joinpath("pr_cpu_lebesgue_20240816_170150.pkl"))
pr_train_benchmark_leb_2k = pr_load_leb['partial_responses_train']
pr_test_benchmark_leb_2k = pr_load_leb['partial_responses_test']
```

Data successfully loaded from
C:\Users\localuser\PRiSM\prism_github\models\pr_cpu_lebesgue_20240816_170150.pkl
Loaded data contains the following keys: x_train, x_test, model,
partial_responses_train, partial_responses_test, method, device
x_train shape: (2000, 11)
x_test shape: (2000, 11)
partial_responses_train shape: (2000, 66)
partial_responses_test shape: (2000, 66)

```
[27]: %reload_ext autoreload

device = 'cpu'

x_train_tensor = torch.tensor(x_train.values, dtype=torch.float32,
    ↪device=device)
x_test_tensor = torch.tensor(x_test.values, dtype=torch.float32, device=device)

pr_train_leb_2k, pr_test_leb_2k, bivariate_inputs_leb_2k = partial_responses(
    x_train_tensor[:2000,:],
    x_test_tensor[:2000:],
    mlp,
    method='lebesgue',
    device=device,
    batch_size=2000);
```

Main compute device: cpu
Max threads: 15.

Univariate 0, batch_start 0	(cpu)
Univariate 1, batch_start 0	(cpu)
Univariate 2, batch_start 0	(cpu)
Univariate 3, batch_start 0	(cpu)
Univariate 4, batch_start 0	(cpu)
Univariate 5, batch_start 0	(cpu)
Univariate 6, batch_start 0	(cpu)
Univariate 7, batch_start 0	(cpu)
Univariate 8, batch_start 0	(cpu)
Univariate 9, batch_start 0	(cpu)
Univariate 10, batch_start 0	(cpu)
Bivariate 0,1, batch_start 0	(cpu)
Bivariate 0,2, batch_start 0	(cpu)
Bivariate 0,3, batch_start 0	(cpu)
Bivariate 0,4, batch_start 0	(cpu)

Bivariate 0,5, batch_start 0 (cpu)
Bivariate 0,6, batch_start 0 (cpu)
Bivariate 0,7, batch_start 0 (cpu)
Bivariate 0,8, batch_start 0 (cpu)
Bivariate 0,9, batch_start 0 (cpu)
Bivariate 0,10, batch_start 0 (cpu)
Bivariate 1,2, batch_start 0 (cpu)
Bivariate 1,3, batch_start 0 (cpu)
Bivariate 1,4, batch_start 0 (cpu)
Bivariate 1,5, batch_start 0 (cpu)
Bivariate 1,6, batch_start 0 (cpu)
Bivariate 1,7, batch_start 0 (cpu)
Bivariate 1,8, batch_start 0 (cpu)
Bivariate 1,9, batch_start 0 (cpu)
Bivariate 1,10, batch_start 0 (cpu)
Bivariate 2,3, batch_start 0 (cpu)
Bivariate 2,4, batch_start 0 (cpu)
Bivariate 2,5, batch_start 0 (cpu)
Bivariate 2,6, batch_start 0 (cpu)
Bivariate 2,7, batch_start 0 (cpu)
Bivariate 2,8, batch_start 0 (cpu)
Bivariate 2,9, batch_start 0 (cpu)
Bivariate 2,10, batch_start 0 (cpu)
Bivariate 3,4, batch_start 0 (cpu)
Bivariate 3,5, batch_start 0 (cpu)
Bivariate 3,6, batch_start 0 (cpu)
Bivariate 3,7, batch_start 0 (cpu)
Bivariate 3,8, batch_start 0 (cpu)
Bivariate 3,9, batch_start 0 (cpu)
Bivariate 3,10, batch_start 0 (cpu)
Bivariate 4,5, batch_start 0 (cpu)
Bivariate 4,6, batch_start 0 (cpu)
Bivariate 4,7, batch_start 0 (cpu)
Bivariate 4,8, batch_start 0 (cpu)
Bivariate 4,9, batch_start 0 (cpu)
Bivariate 4,10, batch_start 0 (cpu)
Bivariate 5,6, batch_start 0 (cpu)
Bivariate 5,7, batch_start 0 (cpu)
Bivariate 5,8, batch_start 0 (cpu)
Bivariate 5,9, batch_start 0 (cpu)
Bivariate 5,10, batch_start 0 (cpu)
Bivariate 6,7, batch_start 0 (cpu)
Bivariate 6,8, batch_start 0 (cpu)
Bivariate 6,9, batch_start 0 (cpu)
Bivariate 6,10, batch_start 0 (cpu)
Bivariate 7,8, batch_start 0 (cpu)
Bivariate 7,9, batch_start 0 (cpu)
Bivariate 7,10, batch_start 0 (cpu)

```

Bivariate 8,9, batch_start 0    (cpu)
Bivariate 8,10, batch_start 0   (cpu)
Bivariate 9,10, batch_start 0   (cpu)
Main compute device: cpu
Max threads: 15.
Univariate 0, batch_start 0     (cpu)
Univariate 1, batch_start 0     (cpu)
Univariate 2, batch_start 0     (cpu)
Univariate 3, batch_start 0     (cpu)
Univariate 4, batch_start 0     (cpu)
Univariate 5, batch_start 0     (cpu)
Univariate 6, batch_start 0     (cpu)
Univariate 7, batch_start 0     (cpu)
Univariate 8, batch_start 0     (cpu)
Univariate 9, batch_start 0     (cpu)
Univariate 10, batch_start 0    (cpu)
Bivariate 0,1, batch_start 0    (cpu)
Bivariate 0,2, batch_start 0    (cpu)
Bivariate 0,3, batch_start 0    (cpu)
Bivariate 0,4, batch_start 0    (cpu)
Bivariate 0,5, batch_start 0    (cpu)
Bivariate 0,6, batch_start 0    (cpu)
Bivariate 0,7, batch_start 0    (cpu)
Bivariate 0,8, batch_start 0    (cpu)
Bivariate 0,9, batch_start 0    (cpu)
Bivariate 0,10, batch_start 0   (cpu)
Bivariate 1,2, batch_start 0    (cpu)
Bivariate 1,3, batch_start 0    (cpu)
Bivariate 1,4, batch_start 0    (cpu)
Bivariate 1,5, batch_start 0    (cpu)
Bivariate 1,6, batch_start 0    (cpu)
Bivariate 1,7, batch_start 0    (cpu)
Bivariate 1,8, batch_start 0    (cpu)
Bivariate 1,9, batch_start 0    (cpu)
Bivariate 1,10, batch_start 0   (cpu)
Bivariate 2,3, batch_start 0    (cpu)
Bivariate 2,4, batch_start 0    (cpu)
Bivariate 2,5, batch_start 0    (cpu)
Bivariate 2,6, batch_start 0    (cpu)
Bivariate 2,7, batch_start 0    (cpu)
Bivariate 2,8, batch_start 0    (cpu)
Bivariate 2,9, batch_start 0    (cpu)
Bivariate 2,10, batch_start 0   (cpu)
Bivariate 3,4, batch_start 0    (cpu)
Bivariate 3,5, batch_start 0    (cpu)
Bivariate 3,6, batch_start 0    (cpu)
Bivariate 3,7, batch_start 0    (cpu)
Bivariate 3,8, batch_start 0    (cpu)

```

```

Bivariate 3,9, batch_start 0    (cpu)
Bivariate 3,10, batch_start 0   (cpu)
Bivariate 4,5, batch_start 0    (cpu)
Bivariate 4,6, batch_start 0    (cpu)
Bivariate 4,7, batch_start 0    (cpu)
Bivariate 4,8, batch_start 0    (cpu)
Bivariate 4,9, batch_start 0    (cpu)
Bivariate 4,10, batch_start 0   (cpu)
Bivariate 5,6, batch_start 0    (cpu)
Bivariate 5,7, batch_start 0    (cpu)
Bivariate 5,8, batch_start 0    (cpu)
Bivariate 5,9, batch_start 0    (cpu)
Bivariate 5,10, batch_start 0   (cpu)
Bivariate 6,7, batch_start 0    (cpu)
Bivariate 6,8, batch_start 0    (cpu)
Bivariate 6,9, batch_start 0    (cpu)
Bivariate 6,10, batch_start 0   (cpu)
Bivariate 7,8, batch_start 0    (cpu)
Bivariate 7,9, batch_start 0    (cpu)
Bivariate 7,10, batch_start 0   (cpu)
Bivariate 8,9, batch_start 0    (cpu)
Bivariate 8,10, batch_start 0   (cpu)
Bivariate 9,10, batch_start 0   (cpu)

```

```
[54]: print_first_responses(pr_train_leb_2k, pr_test_leb_2k)
```

```

First few univariate responses (train):
tensor([[ 0.0222, -0.1418, -0.0857,  0.0479, -0.0491],
        [-0.1942,  0.0582, -0.0524, -0.1643, -0.0491],
        [ 0.2531, -0.0804,  0.0711, -0.0500, -0.0491],
        [ 0.1936,  0.4541,  0.0558,  0.2210, -0.0491],
        [-0.1713, -0.2078, -0.0015, -0.2539, -0.0491]])

First few bivariate responses (train):
tensor([[ 0.0112, -0.0013, -0.0087,  0.0028, -0.0017],
        [ 0.0059, -0.0020, -0.0196, -0.0005, -0.0055],
        [ 0.0115,  0.0144, -0.0073,  0.0020,  0.0064],
        [-0.0576,  0.0138, -0.0346,  0.0026,  0.0041],
        [-0.0169, -0.0038, -0.0341,  0.0002, -0.0059]])

First few univariate responses (test):
tensor([[ -0.5348, -0.3825, -0.4539, -0.4105, -0.4554],
        [-0.5456, -0.4310, -0.4668, -0.3284, -0.4554],
        [-0.5463, -0.3614, -0.4024, -0.4767, -0.4554],
        [-0.5437, -0.4819, -0.2711, -0.5213, -0.4554],
        [-0.1089, -0.4085, -0.3072,  0.0701, -0.4554]])

First few bivariate responses (test):
tensor([[0.4245, 0.4030, 0.4249, 0.4078, 0.4081],
        [0.4099, 0.4904, 0.4209, 0.4079, 0.4074],
        [0.4221, 0.3919, 0.4159, 0.4078, 0.4077],

```

```
[0.4017, 0.3638, 0.4123, 0.4078, 0.4078],
[0.4085, 0.4630, 0.3891, 0.4057, 0.4103]])
```

```
[55]: print_first_responses(pr_train_benchmark_leb_2k, pr_test_benchmark_leb_2k)
```

First few univariate responses (train):

```
[[ 0.02220881 -0.14181936 -0.08567131  0.04791009 -0.04912555]
 [-0.1941973   0.05821919 -0.05236351 -0.16433847 -0.04912555]
 [ 0.25309598 -0.08035314  0.07109129 -0.05002487 -0.04912555]
 [ 0.19358945  0.45411789  0.05583167  0.2210207  -0.04912555]
 [-0.17131197 -0.20779407 -0.00153708 -0.25388396 -0.04912555]]
```

First few bivariate responses (train):

```
[[ 0.01123413 -0.00134256 -0.00868134  0.00279103 -0.00174187]
 [ 0.0058635  -0.0020462  -0.01963957 -0.00053424 -0.00549492]
 [ 0.01152132  0.01435027 -0.00729018  0.00199009  0.00635386]
 [-0.0576371   0.01377874 -0.03455305  0.00261738  0.00406238]
 [-0.0169408  -0.00379623 -0.03409243  0.00015123 -0.00586825]]
```

First few univariate responses (test):

```
[[ -0.53482759 -0.38247073 -0.45387995 -0.41054976 -0.45540726]
 [ -0.54564226 -0.43100011 -0.46683419 -0.32836759 -0.45540726]
 [ -0.54626524 -0.36140859 -0.40236485 -0.47668135 -0.45540726]
 [ -0.54374659 -0.48192441 -0.27110088 -0.52128971 -0.45540726]
 [ -0.10894811 -0.40851128 -0.3072077  0.07007539 -0.45540726]]
```

First few bivariate responses (test):

```
[[0.424481  0.40298256 0.42489856 0.40781677 0.40812531]
 [0.40988296 0.49036708 0.4208529  0.40791523 0.4073635 ]
 [0.42207319 0.39185607 0.41591382 0.40776277 0.40765777]
 [0.40174749 0.36381385 0.41225156 0.4077563  0.40779456]
 [0.4085207  0.46302375 0.38913378 0.40568116 0.41030908]]
```

```
[56]: # To debug differences, check for deviations between benchmark (numpy) and new_
      ↪ (tensor) partial responses
      compare_and_visualize(pr_train_leb_2k, pr_train_benchmark_leb_2k)
```

Data Types:

PyTorch Tensor: torch.float32

NumPy Array: float64

Shape:

PyTorch Tensor: torch.Size([2000, 66])

NumPy Array: (2000, 66)

Basic Statistics:

Mean:

PyTorch Tensor: 0.00013781434972770512

NumPy Array: 0.00013783417153042746

Standard Deviation:

PyTorch Tensor: 0.07101757824420929

NumPy Array: 0.07101731439211904

Min:

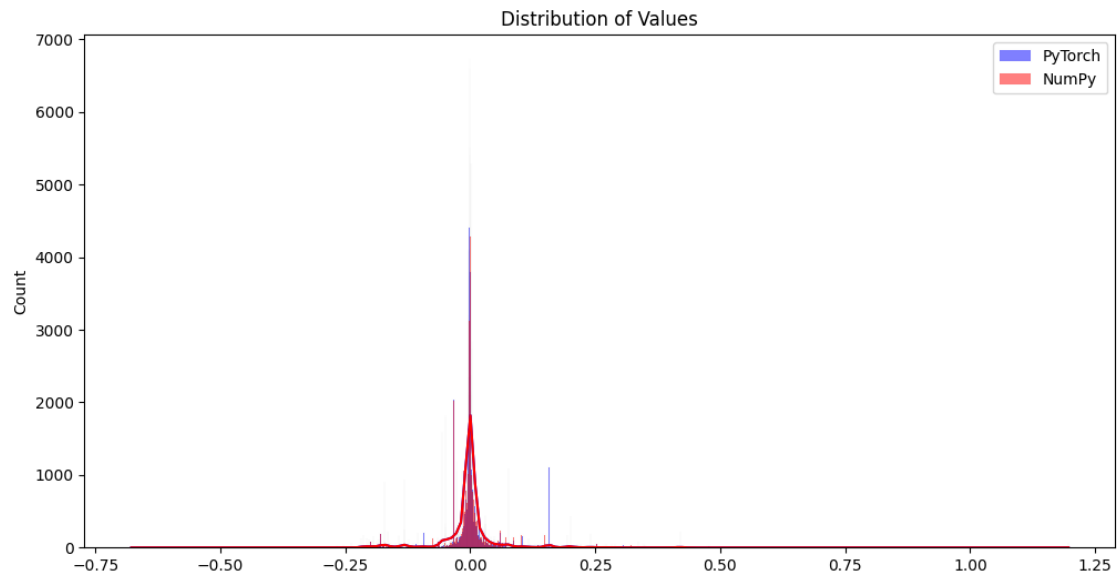
PyTorch Tensor: -0.6779278516769409

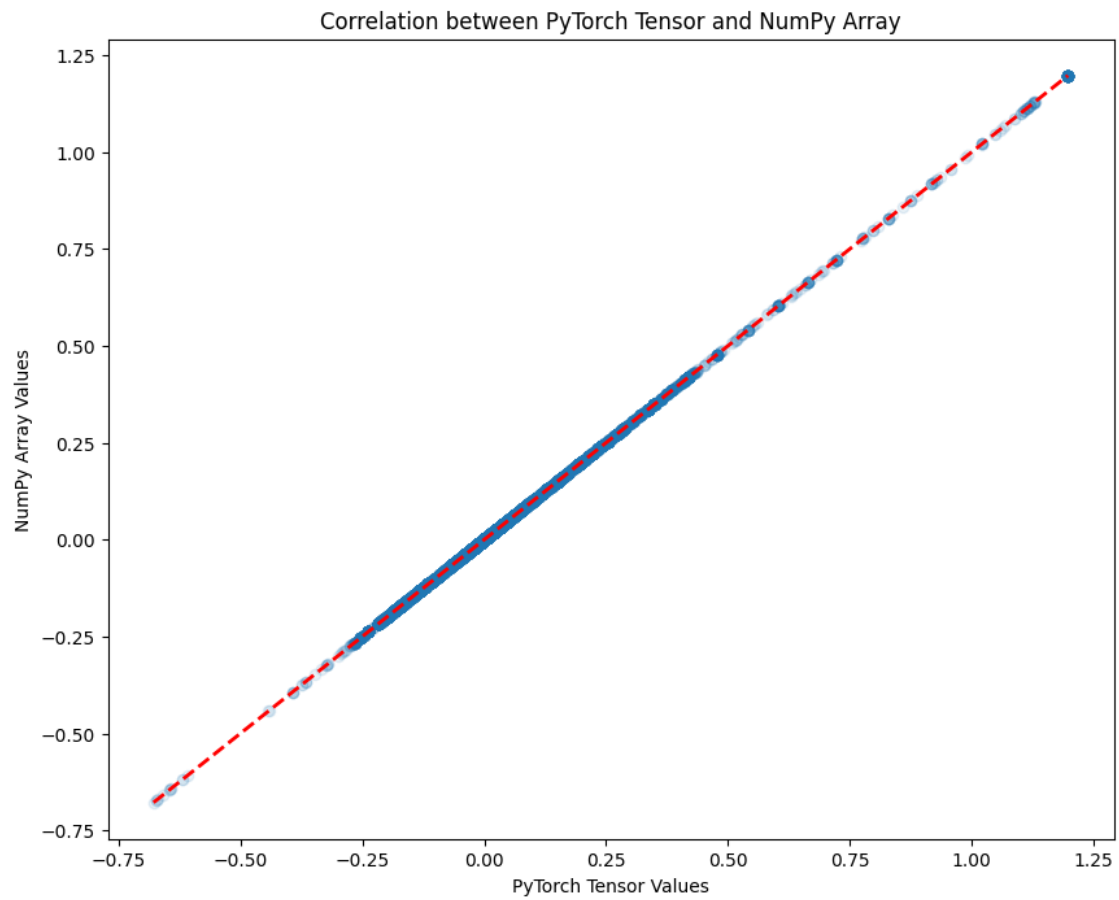
NumPy Array: -0.6779278516769409

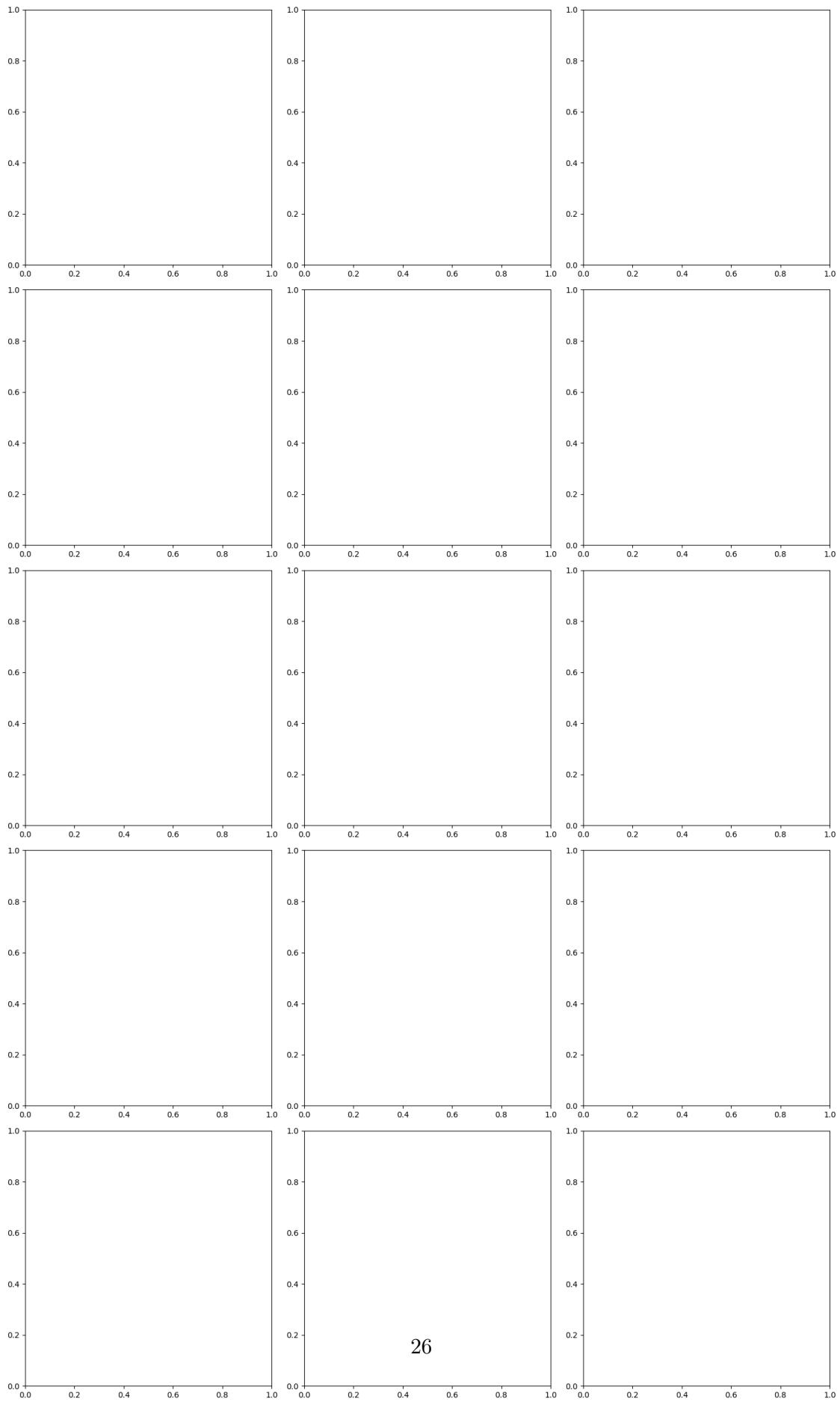
Max:

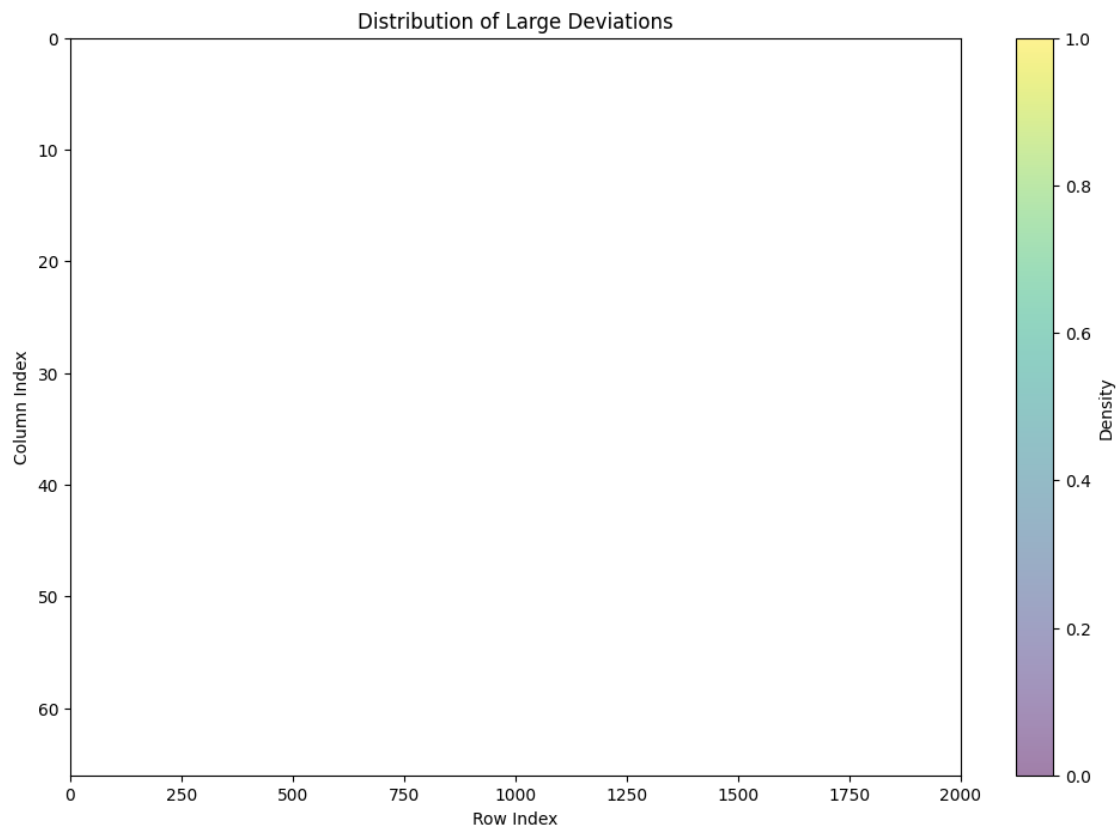
PyTorch Tensor: 1.1969616413116455

NumPy Array: 1.1969616413116455









Number of elements with large deviations: 0
 Percentage of elements with large deviations: 0.00%

```
[28]: %reload_ext autoreload
test_refactored_partial_responses(partial_responses, tensor_input=True, filename=MODELS_DIR.
    ↪joinpath("pr_cpu_lebesgue_20240816_170150.pkl"))
```

```
Main compute device: cpu
Max threads: 15
Univariate 0, batch_start 0      (cpu)
Univariate 0, batch_start 1024   (cpu)
Univariate 1, batch_start 0      (cpu)
Univariate 1, batch_start 1024   (cpu)
Univariate 2, batch_start 0      (cpu)
Univariate 2, batch_start 1024   (cpu)
Univariate 3, batch_start 0      (cpu)
Univariate 3, batch_start 1024   (cpu)
Univariate 4, batch_start 0      (cpu)
Univariate 4, batch_start 1024   (cpu)
```

```

Univariate 5, batch_start 0      (cpu)
Univariate 5, batch_start 1024  (cpu)
Univariate 6, batch_start 0      (cpu)
Univariate 6, batch_start 1024  (cpu)
Univariate 7, batch_start 0      (cpu)
Univariate 7, batch_start 1024  (cpu)
Univariate 8, batch_start 0      (cpu)
Univariate 8, batch_start 1024  (cpu)
Univariate 9, batch_start 0      (cpu)
Univariate 9, batch_start 1024  (cpu)
Univariate 10, batch_start 0     (cpu)
Univariate 10, batch_start 1024 (cpu)
Bivariate 0,1, batch_start 0     (cpu)
Bivariate 0,1, batch_start 1024 (cpu)
Bivariate 0,2, batch_start 0     (cpu)
Bivariate 0,2, batch_start 1024 (cpu)
Bivariate 0,3, batch_start 0     (cpu)
Bivariate 0,3, batch_start 1024 (cpu)
Bivariate 0,4, batch_start 0     (cpu)
Bivariate 0,4, batch_start 1024 (cpu)
Bivariate 0,5, batch_start 0     (cpu)
Bivariate 0,5, batch_start 1024 (cpu)
Bivariate 0,6, batch_start 0     (cpu)
Bivariate 0,6, batch_start 1024 (cpu)
Bivariate 0,7, batch_start 0     (cpu)
Bivariate 0,7, batch_start 1024 (cpu)
Bivariate 0,8, batch_start 0     (cpu)
Bivariate 0,8, batch_start 1024 (cpu)
Bivariate 0,9, batch_start 0     (cpu)
Bivariate 0,9, batch_start 1024 (cpu)
Bivariate 0,10, batch_start 0    (cpu)
Bivariate 0,10, batch_start 1024 (cpu)
Bivariate 1,2, batch_start 0     (cpu)
Bivariate 1,2, batch_start 1024 (cpu)
Bivariate 1,3, batch_start 0     (cpu)
Bivariate 1,3, batch_start 1024 (cpu)
Bivariate 1,4, batch_start 0     (cpu)
Bivariate 1,4, batch_start 1024 (cpu)
Bivariate 1,5, batch_start 0     (cpu)
Bivariate 1,5, batch_start 1024 (cpu)
Bivariate 1,6, batch_start 0     (cpu)
Bivariate 1,6, batch_start 1024 (cpu)
Bivariate 1,7, batch_start 0     (cpu)
Bivariate 1,7, batch_start 1024 (cpu)
Bivariate 1,8, batch_start 0     (cpu)
Bivariate 1,8, batch_start 1024 (cpu)
Bivariate 1,9, batch_start 0     (cpu)
Bivariate 1,9, batch_start 1024 (cpu)

```

Bivariate 1,10, batch_start 0	(cpu)
Bivariate 1,10, batch_start 1024	(cpu)
Bivariate 2,3, batch_start 0	(cpu)
Bivariate 2,3, batch_start 1024	(cpu)
Bivariate 2,4, batch_start 0	(cpu)
Bivariate 2,4, batch_start 1024	(cpu)
Bivariate 2,5, batch_start 0	(cpu)
Bivariate 2,5, batch_start 1024	(cpu)
Bivariate 2,6, batch_start 0	(cpu)
Bivariate 2,6, batch_start 1024	(cpu)
Bivariate 2,7, batch_start 0	(cpu)
Bivariate 2,7, batch_start 1024	(cpu)
Bivariate 2,8, batch_start 0	(cpu)
Bivariate 2,8, batch_start 1024	(cpu)
Bivariate 2,9, batch_start 0	(cpu)
Bivariate 2,9, batch_start 1024	(cpu)
Bivariate 2,10, batch_start 0	(cpu)
Bivariate 2,10, batch_start 1024	(cpu)
Bivariate 3,4, batch_start 0	(cpu)
Bivariate 3,4, batch_start 1024	(cpu)
Bivariate 3,5, batch_start 0	(cpu)
Bivariate 3,5, batch_start 1024	(cpu)
Bivariate 3,6, batch_start 0	(cpu)
Bivariate 3,6, batch_start 1024	(cpu)
Bivariate 3,7, batch_start 0	(cpu)
Bivariate 3,7, batch_start 1024	(cpu)
Bivariate 3,8, batch_start 0	(cpu)
Bivariate 3,8, batch_start 1024	(cpu)
Bivariate 3,9, batch_start 0	(cpu)
Bivariate 3,9, batch_start 1024	(cpu)
Bivariate 3,10, batch_start 0	(cpu)
Bivariate 3,10, batch_start 1024	(cpu)
Bivariate 4,5, batch_start 0	(cpu)
Bivariate 4,5, batch_start 1024	(cpu)
Bivariate 4,6, batch_start 0	(cpu)
Bivariate 4,6, batch_start 1024	(cpu)
Bivariate 4,7, batch_start 0	(cpu)
Bivariate 4,7, batch_start 1024	(cpu)
Bivariate 4,8, batch_start 0	(cpu)
Bivariate 4,8, batch_start 1024	(cpu)
Bivariate 4,9, batch_start 0	(cpu)
Bivariate 4,9, batch_start 1024	(cpu)
Bivariate 4,10, batch_start 0	(cpu)
Bivariate 4,10, batch_start 1024	(cpu)
Bivariate 5,6, batch_start 0	(cpu)
Bivariate 5,6, batch_start 1024	(cpu)
Bivariate 5,7, batch_start 0	(cpu)
Bivariate 5,7, batch_start 1024	(cpu)

```

Bivariate 5,8, batch_start 0      (cpu)
Bivariate 5,8, batch_start 1024 (cpu)
Bivariate 5,9, batch_start 0      (cpu)
Bivariate 5,9, batch_start 1024 (cpu)
Bivariate 5,10, batch_start 0     (cpu)
Bivariate 5,10, batch_start 1024  (cpu)
Bivariate 6,7, batch_start 0      (cpu)
Bivariate 6,7, batch_start 1024 (cpu)
Bivariate 6,8, batch_start 0      (cpu)
Bivariate 6,8, batch_start 1024 (cpu)
Bivariate 6,9, batch_start 0      (cpu)
Bivariate 6,9, batch_start 1024 (cpu)
Bivariate 6,10, batch_start 0     (cpu)
Bivariate 6,10, batch_start 1024  (cpu)
Bivariate 7,8, batch_start 0      (cpu)
Bivariate 7,8, batch_start 1024 (cpu)
Bivariate 7,9, batch_start 0      (cpu)
Bivariate 7,9, batch_start 1024 (cpu)
Bivariate 7,10, batch_start 0     (cpu)
Bivariate 7,10, batch_start 1024  (cpu)
Bivariate 8,9, batch_start 0      (cpu)
Bivariate 8,9, batch_start 1024 (cpu)
Bivariate 8,10, batch_start 0     (cpu)
Bivariate 8,10, batch_start 1024  (cpu)
Bivariate 9,10, batch_start 0     (cpu)
Bivariate 9,10, batch_start 1024  (cpu)
Main compute device: cpu
Max threads: 15
Univariate 0, batch_start 0      (cpu)
Univariate 0, batch_start 1024 (cpu)
Univariate 1, batch_start 0      (cpu)
Univariate 1, batch_start 1024 (cpu)
Univariate 2, batch_start 0      (cpu)
Univariate 2, batch_start 1024 (cpu)
Univariate 3, batch_start 0      (cpu)
Univariate 3, batch_start 1024 (cpu)
Univariate 4, batch_start 0      (cpu)
Univariate 4, batch_start 1024 (cpu)
Univariate 5, batch_start 0      (cpu)
Univariate 5, batch_start 1024 (cpu)
Univariate 6, batch_start 0      (cpu)
Univariate 6, batch_start 1024 (cpu)
Univariate 7, batch_start 0      (cpu)
Univariate 7, batch_start 1024 (cpu)
Univariate 8, batch_start 0      (cpu)
Univariate 8, batch_start 1024 (cpu)
Univariate 9, batch_start 0      (cpu)
Univariate 9, batch_start 1024 (cpu)

```

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Univariate 10, batch_start 0      (cpu)
Univariate 10, batch_start 1024 (cpu)
Bivariate 0,1, batch_start 0      (cpu)
Bivariate 0,1, batch_start 1024 (cpu)
Bivariate 0,2, batch_start 0      (cpu)
Bivariate 0,2, batch_start 1024 (cpu)
Bivariate 0,3, batch_start 0      (cpu)
Bivariate 0,3, batch_start 1024 (cpu)
Bivariate 0,4, batch_start 0      (cpu)
Bivariate 0,4, batch_start 1024 (cpu)
Bivariate 0,5, batch_start 0      (cpu)
Bivariate 0,5, batch_start 1024 (cpu)
Bivariate 0,6, batch_start 0      (cpu)
Bivariate 0,6, batch_start 1024 (cpu)
Bivariate 0,7, batch_start 0      (cpu)
Bivariate 0,7, batch_start 1024 (cpu)
Bivariate 0,8, batch_start 0      (cpu)
Bivariate 0,8, batch_start 1024 (cpu)
Bivariate 0,9, batch_start 0      (cpu)
Bivariate 0,9, batch_start 1024 (cpu)
Bivariate 0,10, batch_start 0     (cpu)
Bivariate 0,10, batch_start 1024  (cpu)
Bivariate 1,2, batch_start 0      (cpu)
Bivariate 1,2, batch_start 1024 (cpu)
Bivariate 1,3, batch_start 0      (cpu)
Bivariate 1,3, batch_start 1024 (cpu)
Bivariate 1,4, batch_start 0      (cpu)
Bivariate 1,4, batch_start 1024 (cpu)
Bivariate 1,5, batch_start 0      (cpu)
Bivariate 1,5, batch_start 1024 (cpu)
Bivariate 1,6, batch_start 0      (cpu)
Bivariate 1,6, batch_start 1024 (cpu)
Bivariate 1,7, batch_start 0      (cpu)
Bivariate 1,7, batch_start 1024 (cpu)
Bivariate 1,8, batch_start 0      (cpu)
Bivariate 1,8, batch_start 1024 (cpu)
Bivariate 1,9, batch_start 0      (cpu)
Bivariate 1,9, batch_start 1024 (cpu)
Bivariate 1,10, batch_start 0     (cpu)
Bivariate 1,10, batch_start 1024  (cpu)
Bivariate 2,3, batch_start 0      (cpu)
Bivariate 2,3, batch_start 1024 (cpu)
Bivariate 2,4, batch_start 0      (cpu)
Bivariate 2,4, batch_start 1024 (cpu)
Bivariate 2,5, batch_start 0      (cpu)
Bivariate 2,5, batch_start 1024 (cpu)
Bivariate 2,6, batch_start 0      (cpu)
Bivariate 2,6, batch_start 1024 (cpu)

```

```

Bivariate 2,7, batch_start 0      (cpu)
Bivariate 2,7, batch_start 1024 (cpu)
Bivariate 2,8, batch_start 0      (cpu)
Bivariate 2,8, batch_start 1024 (cpu)
Bivariate 2,9, batch_start 0      (cpu)
Bivariate 2,9, batch_start 1024 (cpu)
Bivariate 2,10, batch_start 0     (cpu)
Bivariate 2,10, batch_start 1024  (cpu)
Bivariate 3,4, batch_start 0      (cpu)
Bivariate 3,4, batch_start 1024 (cpu)
Bivariate 3,5, batch_start 0      (cpu)
Bivariate 3,5, batch_start 1024 (cpu)
Bivariate 3,6, batch_start 0      (cpu)
Bivariate 3,6, batch_start 1024 (cpu)
Bivariate 3,7, batch_start 0      (cpu)
Bivariate 3,7, batch_start 1024 (cpu)
Bivariate 3,8, batch_start 0      (cpu)
Bivariate 3,8, batch_start 1024 (cpu)
Bivariate 3,9, batch_start 0      (cpu)
Bivariate 3,9, batch_start 1024 (cpu)
Bivariate 3,10, batch_start 0     (cpu)
Bivariate 3,10, batch_start 1024  (cpu)
Bivariate 4,5, batch_start 0      (cpu)
Bivariate 4,5, batch_start 1024 (cpu)
Bivariate 4,6, batch_start 0      (cpu)
Bivariate 4,6, batch_start 1024 (cpu)
Bivariate 4,7, batch_start 0      (cpu)
Bivariate 4,7, batch_start 1024 (cpu)
Bivariate 4,8, batch_start 0      (cpu)
Bivariate 4,8, batch_start 1024 (cpu)
Bivariate 4,9, batch_start 0      (cpu)
Bivariate 4,9, batch_start 1024 (cpu)
Bivariate 4,10, batch_start 0     (cpu)
Bivariate 4,10, batch_start 1024  (cpu)
Bivariate 5,6, batch_start 0      (cpu)
Bivariate 5,6, batch_start 1024 (cpu)
Bivariate 5,7, batch_start 0      (cpu)
Bivariate 5,7, batch_start 1024 (cpu)
Bivariate 5,8, batch_start 0      (cpu)
Bivariate 5,8, batch_start 1024 (cpu)
Bivariate 5,9, batch_start 0      (cpu)
Bivariate 5,9, batch_start 1024 (cpu)
Bivariate 5,10, batch_start 0     (cpu)
Bivariate 5,10, batch_start 1024  (cpu)
Bivariate 6,7, batch_start 0      (cpu)
Bivariate 6,7, batch_start 1024 (cpu)
Bivariate 6,8, batch_start 0      (cpu)
Bivariate 6,8, batch_start 1024 (cpu)

```



```

Bivariate 6,9, batch_start 0      (cpu)
Bivariate 6,9, batch_start 1024 (cpu)
Bivariate 6,10, batch_start 0     (cpu)
Bivariate 6,10, batch_start 1024 (cpu)
Bivariate 7,8, batch_start 0      (cpu)
Bivariate 7,8, batch_start 1024 (cpu)
Bivariate 7,9, batch_start 0      (cpu)
Bivariate 7,9, batch_start 1024 (cpu)
Bivariate 7,10, batch_start 0     (cpu)
Bivariate 7,10, batch_start 1024 (cpu)
Bivariate 8,9, batch_start 0      (cpu)
Bivariate 8,9, batch_start 1024 (cpu)
Bivariate 8,10, batch_start 0     (cpu)
Bivariate 8,10, batch_start 1024 (cpu)
Bivariate 9,10, batch_start 0     (cpu)
Bivariate 9,10, batch_start 1024 (cpu)
Refactored function produces different results from the original implementation.
atol = 1e-07, rtol = 1e-05
Differences found in partial_responses_train
Max absolute difference: 1.1080410331487656e-06
Max relative difference: 8.640698481882303e-05
Differences found in partial_responses_test
Max absolute difference: 1.2814998626708984e-06
Max relative difference: 2.03014604464616e-05

```

[28]: False

5 Test lebesgue GPU

```
[22]: print(torch.cuda.is_available())
```

True

```

[23]: # View new lebesgue results
      %reload_ext autoreload

      device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
      mlp = mlp.to(device)

      x_train_tensor = torch.tensor(x_train.values, dtype=torch.float32,
      ↪device=device)
      x_test_tensor = torch.tensor(x_test.values, dtype=torch.float32, device=device)

      pr_train_leb, pr_test_leb, bivariate_inputs_leb =
      ↪partial_responses(x_train_tensor[:500,:], x_test_tensor[:500,:
      ↪], mlp, method='lebesgue', device=device)

```

```
[24]: print(f"First few univariate responses (train): {pr_train_leb[:5, :5]}")
print(f"First few bivariate responses (train): {pr_train_leb[:5, 11:16]}")
print(f"First few univariate responses (test): {pr_test_leb[:5, :5]}")
print(f"First few bivariate responses (test): {pr_test_leb[:5, 11:16]}")
```

```
First few univariate responses (train): tensor([[ 0.0228, -0.1354, -0.0824,
0.0514, -0.0488],
        [-0.1893,  0.0641, -0.0497, -0.1613, -0.0488],
        [ 0.2478, -0.0740,  0.0725, -0.0465, -0.0488],
        [ 0.1899,  0.4599,  0.0573,  0.2237, -0.0488],
        [-0.1667, -0.2017,  0.0005, -0.2523, -0.0488]], device='cuda:0')
First few bivariate responses (train): tensor([[ 0.0098, -0.0015, -0.0078,
0.0016, -0.0016],
        [ 0.0045, -0.0035, -0.0181, -0.0017, -0.0062],
        [ 0.0097,  0.0145, -0.0056,  0.0005,  0.0098],
        [-0.0577,  0.0139, -0.0323,  0.0012,  0.0065],
        [-0.0162, -0.0050, -0.0325, -0.0010, -0.0069]], device='cuda:0')
First few univariate responses (test): tensor([[ -0.5321, -0.3919, -0.4584,
-0.4138, -0.4498],
        [-0.5453, -0.4386, -0.4736, -0.3323, -0.4498],
        [-0.5446, -0.3714, -0.4067, -0.4793, -0.4498],
        [-0.5417, -0.4862, -0.2747, -0.5234, -0.4498],
        [-0.1191, -0.4170, -0.3111,  0.0644, -0.4498]], device='cuda:0')
First few bivariate responses (test): tensor([[ 0.4250, 0.4071, 0.4276, 0.4116,
0.4124],
        [0.4116, 0.4927, 0.4240, 0.4113, 0.4116],
        [0.4234, 0.3962, 0.4183, 0.4113, 0.4119],
        [0.4022, 0.3685, 0.4144, 0.4114, 0.4121],
        [0.4162, 0.4682, 0.3921, 0.4082, 0.4140]], device='cuda:0')
```

```
[25]: print(f"First few univariate responses (train): {pr_train_benchmark_leb[:5, :
↵5]}")
print(f"First few bivariate responses (train): {pr_train_benchmark_leb[:5, 11:
↵16]}")
print(f"First few univariate responses (test): {pr_test_benchmark_leb[:5, :5]}")
print(f"First few bivariate responses (test): {pr_test_benchmark_leb[:5, 11:
↵16]}")
```

```
First few univariate responses (train): [[ 0.02279246 -0.13538277 -0.08238709
0.05137408 -0.04876649]
        [-0.18926609  0.06411755 -0.04966962 -0.16132271 -0.04876649]
        [ 0.24780893 -0.07403123  0.07245362 -0.04652226 -0.04876649]
        [ 0.1899035  0.45994484  0.0572989  0.2237339  -0.04876649]
        [-0.1667074  -0.20167506  0.00046611 -0.25231373 -0.04876649]]
First few bivariate responses (train): [[ 0.00975614 -0.0014504  -0.00784853
0.0015941  -0.00162662]
        [ 0.00449143 -0.00350396 -0.01813054 -0.00167009 -0.00617562]
        [ 0.00969791  0.01454909 -0.00556087  0.0005069  0.00979852]
```

```

[-0.05770344  0.01386476 -0.0322869   0.00121743  0.00653278]
[-0.0161572  -0.00503564 -0.03246517 -0.00096593 -0.00687471]]
First few univariate responses (test): [[-0.53211224 -0.39187014 -0.45838583
-0.41380584 -0.44975626]
[-0.54525983 -0.43861377 -0.47356451 -0.33227766 -0.44975626]
[-0.54459131 -0.37141097 -0.40668476 -0.47929466 -0.44975626]
[-0.54169858 -0.48624599 -0.27472937 -0.52340114 -0.44975626]
[-0.11911523 -0.41703665 -0.31105554  0.06441379 -0.44975626]]
First few bivariate responses (test): [[0.42502758 0.40714574 0.42756259
0.41155931 0.41243818]
[0.41156316 0.49267107 0.42399994 0.41126719 0.41159272]
[0.42337883 0.39622626 0.41832441 0.41132978 0.41194433]
[0.40220404 0.36848888 0.41442394 0.41138661 0.41209343]
[0.41623333 0.46823516 0.39210296 0.40820089 0.41403693]]

```

5.1 Test Lebesgue GPU full dataset (batching)

```

[26]: import torch

print(f"GPU available: {torch.cuda.is_available()}")
if torch.cuda.is_available():
    print(f"GPU memory allocated: {torch.cuda.memory_allocated() / 1e9:.2f} GB")
    print(f"GPU memory cached: {torch.cuda.memory_reserved() / 1e9:.2f} GB")

GPU available: True
GPU memory allocated: 0.01 GB
GPU memory cached: 0.05 GB

[27]: # View new lebesgue results
%reload_ext autoreload

device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
mlp = mlp.to(device)

x_train_tensor = torch.tensor(x_train.values, dtype=torch.float32,
    ↪device=device)
x_test_tensor = torch.tensor(x_test.values, dtype=torch.float32, device=device)

pr_train_leb_full, pr_test_leb_full, bivariate_inputs_leb_full =
    ↪partial_responses(x_train_tensor, x_test_tensor, mlp, method='lebesgue', device=device)

[29]: pr_train_leb_full.shape

[29]: torch.Size([31315, 66])

[30]: print(f"First few univariate responses (train): {pr_train_leb_full[:5, :5]}")
print(f"First few bivariate responses (train): {pr_train_leb_full[:5, 11:16]}")
print(f"First few univariate responses (test): {pr_test_leb_full[:5, :5]}")

```

```
print(f"First few bivariate responses (test): {pr_test_leb_full[:5, 11:16]}")
```

```
First few univariate responses (train): tensor([[ 0.0137, -0.1264, -0.0871,
 0.0716, -0.0474],
        [-0.1790,  0.0454, -0.0552, -0.1678, -0.0474],
        [ 0.2427, -0.0768,  0.0676, -0.0361, -0.0474],
        [ 0.1822,  0.4337,  0.0522,  0.2546, -0.0474],
        [-0.1620, -0.1692, -0.0053, -0.2804, -0.0474]], device='cuda:0')
First few bivariate responses (train): tensor([[ 0.0114, -0.0006, -0.0085,
 0.0014, -0.0017],
        [ 0.0074, -0.0019, -0.0155, -0.0013, -0.0037],
        [ 0.0107,  0.0210, -0.0091,  0.0003,  0.0049],
        [-0.0525,  0.0187, -0.0330,  0.0011,  0.0030],
        [-0.0157, -0.0054, -0.0326, -0.0008, -0.0043]], device='cuda:0')
First few univariate responses (test): tensor([[ -0.4581, -0.3130, -0.3826,
 -0.3431, -0.3830],
        [-0.4712, -0.3577, -0.3691, -0.2595, -0.3830],
        [-0.4707, -0.2934, -0.3337, -0.4108, -0.3830],
        [-0.4678, -0.4023, -0.2068, -0.4568, -0.3830],
        [-0.0405, -0.3371, -0.2419,  0.1371, -0.3830]], device='cuda:0')
First few bivariate responses (test): tensor([[ 0.3591, 0.3329, 0.3564, 0.3409,
 0.3417],
        [0.3438, 0.4213, 0.3521, 0.3409, 0.3403],
        [0.3559, 0.3217, 0.3483, 0.3408, 0.3409],
        [0.3370, 0.2921, 0.3452, 0.3408, 0.3412],
        [0.3395, 0.4006, 0.3300, 0.3379, 0.3427]], device='cuda:0')
```

```
[66]: # Saving
filename_pr_leb_full = MODELS_DIR.joinpath('partial_responses_data_leb_all.pkl')

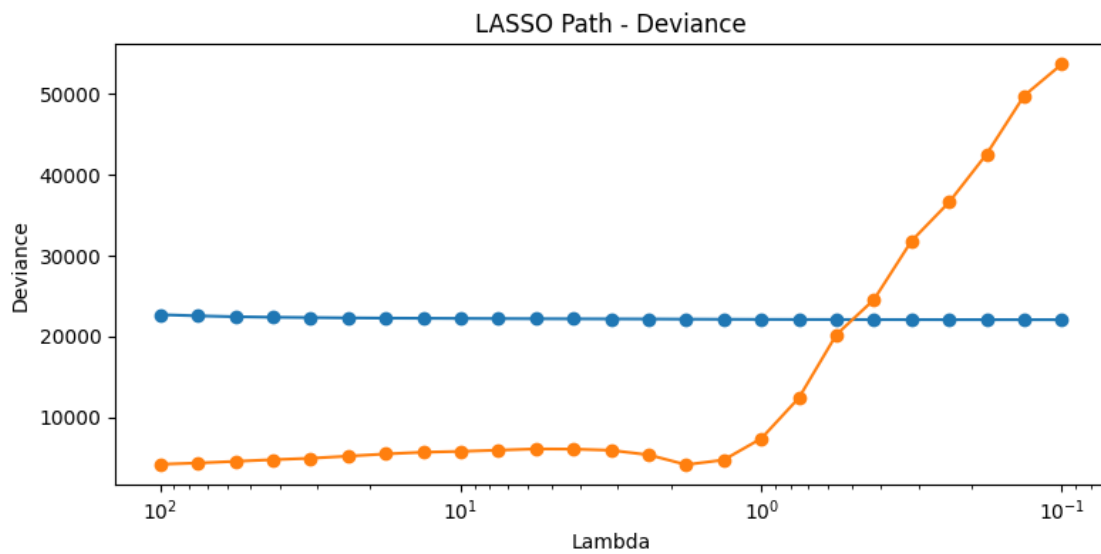
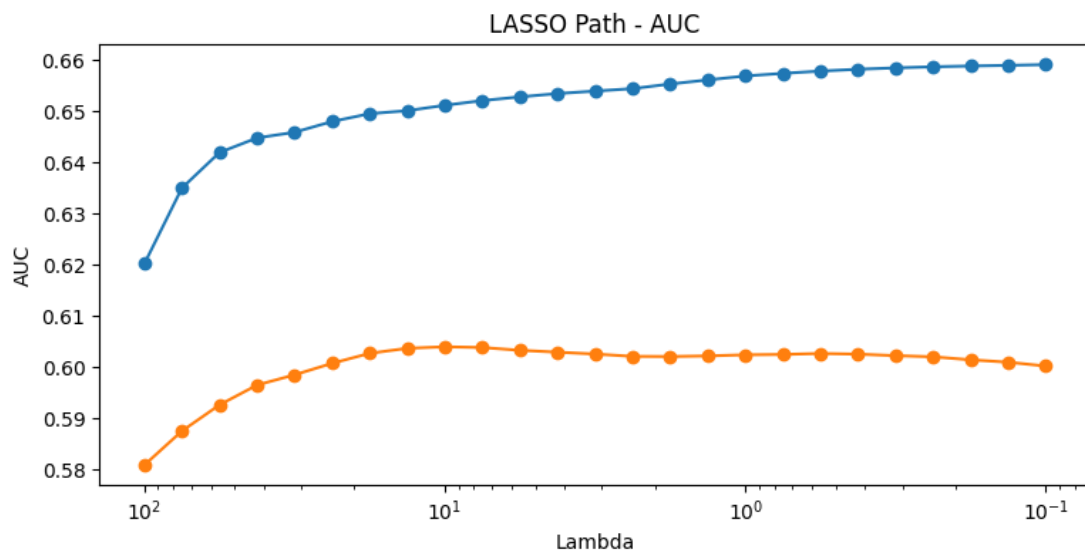
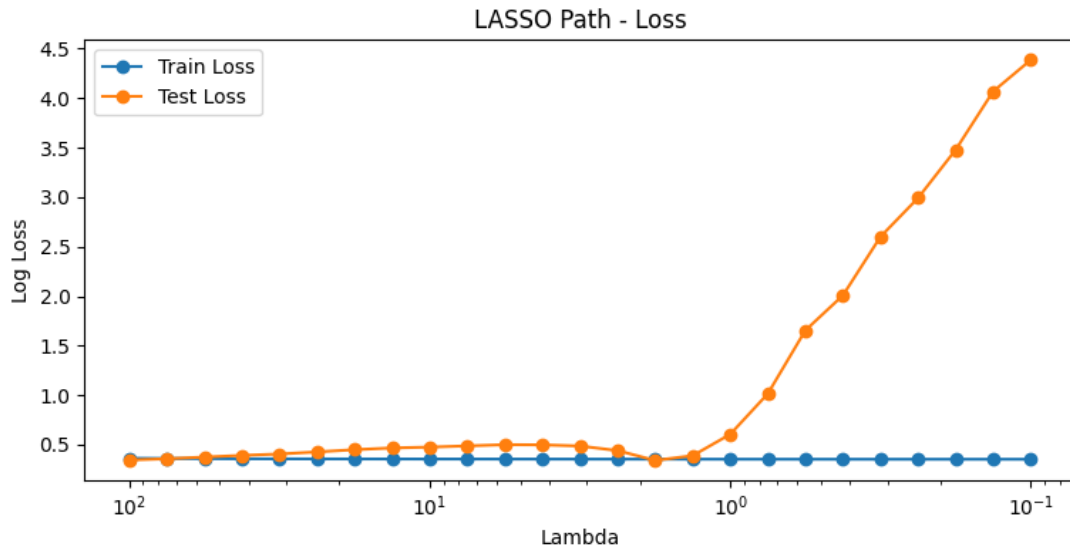
save_partial_responses_results(x_train_tensor, x_test_tensor, mlp,
    ↪ pr_train_leb_full, pr_test_leb_full, bivariate_inputs_leb_full, 'lebesgue',
    ↪ device, filename=filename_pr_leb_full)
```

Results and arguments saved to
C:\Users\localuser\PRiSM\prism_github\models\partial_responses_data_leb_all.pkl

5.2 Lebesgue results nomogram

```
[37]: lasso_leb_full = lasso_results = lasso(
    pr_train_leb_full,
    pr_test_leb_full,
    y_train,
    y_test,
    bivariate_inputs_leb_full,
    feature_names=feature_names,
    nlambdas=25,
    min_lambda=0.1,
```

```
max_lambda=100,  
tol=1e-4,  
batch_size=2  
)
```



<IPython.core.display.HTML object>

```
[40]: lasso_leb_full.select_lambda(8)
lasso_leb_full.plot_lambda_path()
lasso_leb_full.plot_feature_importance()
```

